AI Screening System Testing Report – Zecpath AI

Objective

To validate the accuracy, reliability, and real-world performance of the AI screening system and optimize weak areas.

Test Setup

Parameter	Value
Total Candidates	50
Questions per Candidate	7
Total Responses	350
Roles Tested	Backend, Frontend, HR
Evaluation Method	AI vs Human Reviewer

Test Methodology

1. Simulated AI screening calls conducted
2. Answers processed via:
    - STT Engine
    - Intent Engine
    - Scoring Engine
    - Behavior Analysis
3. Compared AI decisions with human HR evaluation
4. Logged mismatches and error cases

Accuracy Metrics

Metric	Before	After Optimization
Intent Accuracy	78%	88%
Scoring Accuracy	72%	84%

Metric	Before	After Optimization
Overall System Accuracy	75%	86%
False Rejection Rate	18%	9%

Error Analysis

Issue Type	% Cases	Fix Applied
Misclassified Intent	12%	Improved keyword mapping
Short Answer Misjudged	10%	Adjusted clarity scoring
Silence Misinterpreted	8%	Better silence detection
Over-strict scoring	15%	Threshold tuning

Final Result

- System Accuracy Improved to 86%
- False Rejections Reduced by 50%

Improvements Made

Area	Improvement
Intent Engine	Better keyword scoring
Scoring Engine	Balanced thresholds
Conversation Flow	Adaptive retry
Error Handling	Better silence detection

Before vs After Example

Before Optimization

```
{
  "score": 48,
  "decision": "Reject"
}
```